



Faculty of Applied Sciences
University of Sri Jayewardenepura, Sri Lanka

ANALYZING THE IMPACT OF STUDY HABITS, ATTENDANCE, AND DIGITAL TOOL USAGE ON ACADEMIC PERFORMANCE

STA 330 2.0 Data Analysis and Preparation of Reports

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Abstract

This study investigates the relationship between key behavioral and technological factors study hours, class attendance, class engagement, and digital learning practices and academic performance among third-year undergraduate students of the Faculty of Applied Science, University of Sri Jayewardenepura. The aim is to identify the most significant predictors of academic achievement and provide empirical insight into the study habits that contribute to higher performance outcomes in higher education. Data for this analysis were collected through a Google Form administered to undergraduate students across various disciplines at a university.

The analysis employs a multiple linear regression model, using students' self-reported GPA as the dependent variable. Independent variables include gender, weekly study hours, percentage of lecture and tutorial attendance, engagement level, frequency of digital and AI tool usage, whether or not a fixed study schedule is used, and academic satisfaction.

These findings suggest that learning behaviors such as consistent digital tool usage, gender, using a fixed study schedule, and active class participation remain critical drivers of academic success, while study time, high attendance and academic satisfaction alone may not substantially enhance academic outcomes without corresponding engagement and effort.

Acknowledgement

As third-year students from the Department of Statistics at the University of Sri Jayewardenepura, we are pleased to present this final report for our “Data Analysis and Preparation” course. This project has been a valuable learning experience, significantly contributing to our academic and professional development.

First and foremost, we extend our heartfelt gratitude to our course lecturer, Mr. P. Dias. His consistent support, insightful guidance, and clear instruction were instrumental to the success of this project. We are truly grateful for the knowledge he imparted and the encouragement he provided throughout the course.

We are also thankful to the Department of Statistics for offering the STA 330.2 “Data Analysis and Preparation” course as part of our academic program. This course equipped us with essential skills in data handling, statistical analysis, and interpretation—skills that are vital for both academic and applied research.

Our sincere appreciation goes to all our group members for their dedication, collaboration, and mutual support, as well as to our families and friends for their encouragement and understanding. The diverse ideas, shared responsibilities, and strong team spirit made this project both enriching and enjoyable.

In conclusion, this research project has significantly enhanced our practical understanding of data analysis. The challenges we faced, the skills we developed, and the lessons we learned will undoubtedly benefit us in our future academic and professional endeavors. We hope that the knowledge gained and experiences shared during this project will continue to guide and inspire us in the years ahead.

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1. Introduction

1.1 Background of the study

Academic performance is a key indicator of the theoretical knowledge of an undergraduate and is influenced by multiple factors, including study habits, class attendance, and the use of digital learning tools. Understanding the relationship between these factors and academic performance is essential for improving student outcomes and designing effective educational strategies.

Effective study habits, such as regular study schedules, revision practices, and self-study, have some kind of relationship with higher academic achievement. Research suggests that students who allocate consistent study hours and engage in active learning techniques tend to perform better in exams and coursework (Nonis & Hudson, 2010). In addition to that, class attendance plays a significant role in academic performance, since it provides direct contact with course content and instructor guidance. Studies indicate a strong positive correlation between lecture attendance and GPA, highlighting the importance of active participation in formal education settings (Schmidt, 2017).

The rapid development of educational technology has affected various ways to the academic work of undergraduates. Online platforms, digital libraries, AI-powered study elements, and chatbots provide some additional learning support beyond the traditional classroom environment. While these tools enhance accessibility and flexibility, their effectiveness in improving academic performance depends on students' engagement levels and proper utilization (Selwyn, 2016).

Although those factors have some kind of effect on academic performance, studying their influence on academic performance with data which are collected from 3rd-year undergraduates of University of Sri Jayewardenepura is important to understanding trends and behaviors unique to this group. This study conducts to achieve that and quantify the impact of study habits, attendance, and digital tool usage on undergraduates' academic performance. The findings will provide valuable insights to undergraduates for their academic success, educators and policymakers develop more effective learning strategies.

1.2 Objective of Study

This study aims to examine the key factors influencing academic performance among university students by analyzing how study habits, attendance, and digital learning tools contribute to students' GPA. The main objective is to identify significant predictors of academic performance by assessing various study-related behaviors and external factors that may have a measurable impact on students' success. In addition to that, this study aims to explore several related aspects that may influence student success. It will examine the role of study habits and attendance by analyzing how factors such as study duration,

revision frequency, and structured study schedules contribute to academic outcomes. Furthermore, the study will investigate the impact of lecture and tutorial attendance, assessing whether higher attendance rates and active class engagement lead to better academic performance. Another important focus is the role of digital learning tools, including e-books, online courses, and AI-powered educational resources, in enhancing learning and comprehension. The study will compare students who frequently use these digital platforms with those who rely on traditional study methods to determine the effectiveness of technology in education. Additionally, it will explore whether technology can compensate for lower study hours or reduced attendance. Lastly, based on the findings, the study will provide recommendations for students and institutions to improve study strategies, encourage effective resource utilization, and support the integration of digital tools into academic environments.

1.3 Limitation of Study

This study has a few limitations. First, it focuses only on 3rd-year students of FAS USJ, so the results may not apply to students from other faculties, universities, or academic years. Second, the data is collected through a self-reported questionnaire, which means students might not always provide completely accurate answers about their study habits, attendance, or use of digital tools. Third, the study does not consider other important factors like family income, mental health, personal responsibility, or the quality of teaching, which could also affect academic performance. Additionally, since this research uses only numbers and statistics (quantitative method), it may not give a deep understanding of students' real experiences compared to interviews or a mix of methods. Finally, limited time and resources might reduce the number of students who can participate, which could affect the accuracy of the results.

1.4 Description of Data

This study was done using 83 observations and 07 predictor to examine the key factors influencing academic performance among university students by analyzing how study habits, attendance, and digital learning tools contribute to students' GPA.

Variable	Description
Gender	Gender(Male/Female)
Study hours	Average study hours (Excluding lectures & tutorials)
Attendance	Average percentage of lecture and tutorials/practical attendance per week (0-100%)
EngagedClass	Engaged level during lectures/tutorials(5 point liker scale)
digital tools usage	How frequently are digital tools used (E-books, Online courses, Educational apps) for studying (Never, Occasionally, Weekly, Daily)
Academic satisfaction	Satisfaction level of academic performance (5-point Likert scale)
Fixed study schedule	Do you follow a fixed study schedule?(Yes/No)

2. Literature Review

There is an immense amount of researches done on the factors affecting undergraduates' performance all around the world due to its critical effect on the job market both directly and indirectly. Even though the earlier studies done by Pilcher & Walters(1997) show that there is a relation between length of sleep and academic performance of college students, Nonis and Hudson (2006) found that the amount of time spent studying or at work had no direct influence on academic performance. Rahimpour and Magsoudpour (2011) studied teacher-students' interactions in task-based vs. formfocused instruction and found that students are motivated towards completion of a task when it is different and stimulating and Clay and Breslow (2006) found that the factor which motivates the students to attend classes is the way of teaching of the content using active learning approaches by the lecturer even if the topic under discussion is not interesting. Also, Romer (1993) and Newman- Ford et al. (2009) state that the major reasons given by students for poor attendance were assessment pressures, timing of lectures, poor delivery of lectures and working commitments and attendance affects performance of students. according to Farooq et al. (2011) SocioEconomic Status (SES) and education of parents have a significant role in the overall academic achievement of the students, and he also found that girl students perform better than the male students concluding there is an effect of gender on students' performance. This support by the findings of Everett and Robins (1991), Abbott-chapman et al. (1997), Win and Miller (2005). But Myatt and Waddell (1990), Mandal et al. (2012) reported that male students obtained better performance than their female students thus contradicting those findings. And by a study done by Osaikhiuwa(2014) found that factors such as unfavorable learning environment, inadequate water supply and insufficient library facilities have no effect while factors escrowed lectures room, breakdowns of electricity supply, continuous strikes have an effect on the performance. Rossi (2017) examined the relationship between academic performance, employment, and parentalsocioeconomic status among evening university students, addressing a gap in research primarily focused on daytime students. Using survey data from 808 students at Universidad Andres Bello and employing OLS and logistic regression models, the study found no statistically significant relationship between academic performance and both employment and parental socioeconomic status. This contrasts with findings for daytime students, suggesting that evening students' financial independence and work experience may influence these outcomes. Dissanayaka et al. (2021) examined the impact of technology readiness, internet self-efficacy, and computing experience on the employability of professional accounting students in Sri Lanka. Using empirical data and regression analysis, the study found significant positive relationships between these three factors and employability. The findings highlight the need for integrating IT education into accounting curricula to enhance students' job readiness.

3. Methodology

3.1. Data Collection

Data for this study were collected using a structured Google Form, which was shared mainly through WhatsApp. This method was chosen as a quick and efficient way to reach the 3rd-year undergraduates at the Faculty of Applied Sciences, University of Sri Jayewardenepura.

Before full-scale distribution, a pilot survey was conducted to identify and correct any errors in the form, ensuring the clarity and quality of the data collected.

To ensure comprehensive coverage, follow-up methods such as telephone conversations and emails were also employed to reach individuals who did not initially respond. Given the large target population, a non-random sampling strategy was used, and data were collected from 83 students, balancing practicality with data representativeness. To maintain data privacy, all sensitive information collected through the survey was handled securely, with assurances that it would not be shared with any third parties.

3.2. Data Entry and Cleaning

After the responses were collected, the data were entered into Minitab for further analysis. Then first check for values that are not in the expected range under each variable. If there are some values, then remove those values and treat them as missing values. Then, all variables were checked for proper coding, with ordinal scales assigned numerical values to maintain the order of categories, ensuring consistent interpretation during analysis. Outliers for quantitative data were identified using box plots. To examine quantitative variables by qualitative groups, box plots were also used (GPA by Gender, GPA by Digital Tool Usage, GPA by Engaged Class, GPA by Academic Satisfaction, and GPA by fixed study schedule). For qualitative variables, bar charts and frequency tables were used to identify unusual observations.

3.3 Data Analysis

3.3.1 Descriptive Statistics

Categorical and ordinal data, including gender, digital tool usage, and fixed study schedules, were analyzed using pie charts and frequency distribution. Relationships between pairs of qualitative variables were explored using two-way frequency tables.

For continuous variables, such as study hours and attendance percentage, measures like mean, median, and standard deviation were computed. If the data was found to be symmetric and free of significant outliers (as identified using dot plots and box plots), the mean and standard deviation were used. In cases where the data was skewed or contained outliers, the median and interquartile range (IQR) were used instead.

3.3.2 Correlation Analysis

3.3.2.1 Qualitative variables with GPA

To assess the link between Gender and GPA, a Two-Sample T-Test was chosen because it compares the average GPA of two independent groups, males and females. Before using this test, the data was checked for normality to ensure it met the required assumptions. If the GPA data for both groups was normally distributed, an F-Test was then used to verify if the variances were similar, allowing for a more accurate comparison. If the normality assumption was not satisfied, the Mann-Whitney U Test was used instead, as it is more suitable for non-normal data.

To analyze whether having a Fixed Study Schedule affects GPA, a Two-Sample T-Test was also used. This test was chosen because it compares the average GPA of two independent groups of students with and without a fixed study schedule. Before conducting the T-Test, the normality of each group's GPA was checked using probability plots. If both groups met the normality requirement, an F-Test was used to determine whether their variances were similar, ensuring the accuracy of the T-Test.

The Kruskal-Wallis Test was chosen to evaluate the relationship between GPA and Digital Tool Usage (Daily, Never, Occasionally, Weekly). This non-parametric test is suitable because the data does not meet the assumption of equal variances required for an ANOVA test. It compares the medians across multiple groups to determine if at least one group differs significantly, making it ideal for ordinal or non-normally distributed data.

The Kruskal-Wallis Test was used to assess the relationship between GPA and Engagement in Class, as the engagement levels are ordinal and do not necessarily follow ANOVA assumptions. This test was chosen because it does not assume normality and is robust to the presence of outliers, making it a reliable choice for comparing medians across multiple engagement levels.

Similarly, the Kruskal-Wallis Test was applied to evaluate the link between GPA and Academic Satisfaction. This non-parametric method was selected due to the ordinal nature of the academic satisfaction data and the potential for non-normal distributions, which violate the assumptions of more conventional parametric tests like ANOVA.

3.3.2.2 Quantitative variables with GPA

This approach is appropriate for continuous, normally distributed data where the goal is to measure the strength and direction of a linear relationship between two variables. The Pearson method was chosen over the Spearman rank correlation because both variables are continuous and were expected to have a roughly linear relationship.

Another Pearson correlation analysis was performed to explore the connection between GPA and Attendance. This method was selected for its ability to quantify the linear relationship between two continuous variables, providing a straightforward measure of association that is easy to interpret.

3.3.3 Regression Analysis

To assess the combined effects of multiple qualitative predictors on GPA, a regression model was developed using Gender, Digital Tool Usage, Fixed Study Schedule, and their interactions. Including interaction terms allows the model to capture more complex relationships, where the impact of one factor depends on the level of another. For example, the effect of Digital Tool Usage on GPA might differ based on Gender. The model included the following main effects and interactions:

Quantitative predictors, including Study Hours, Attendance, Engaged Class, and Academic Satisfaction, were first evaluated using Pearson correlation to assess their individual relationships with GPA. This step provided initial insights into the strength and direction of these associations before they were included in the full regression model.

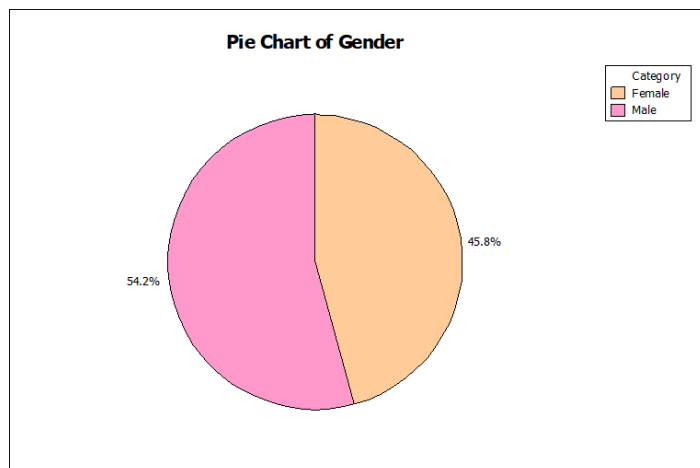
Based on the Minitab output, the final regression model included both qualitative main effects, quantitative main effects, and interaction terms to capture the combined influence of these factors on GPA.

4. Analysis of Data

4.1 Composition of the Sample

4.1.1 Composition of the sample with respect to Gender

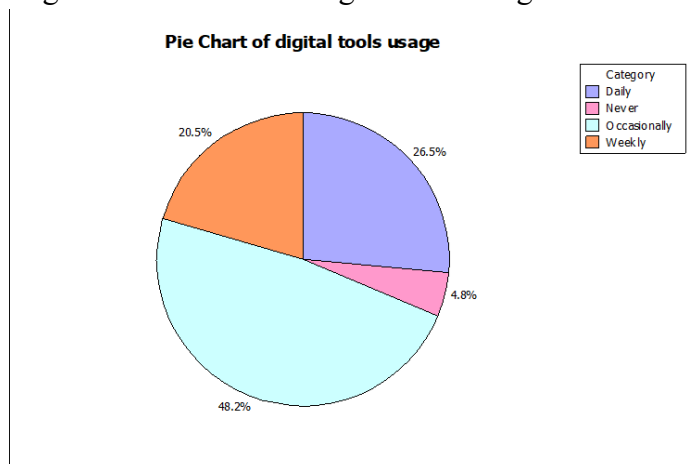
Figure 01- Pie chart of gender



According to Figure 1, there is a higher percentage of males than females, which is 54% for males.

4.1.2 Composition of the sample with respect to digital tools usage

Figure 02- Pie chart of digital tools usage



According to Figure 2, digital tool usage is highest for occasional use and lowest for never use.

4.1.3 Composition of the sample with respect to fixed study schedule

Figure 03- Pie chart of fixed study schedule

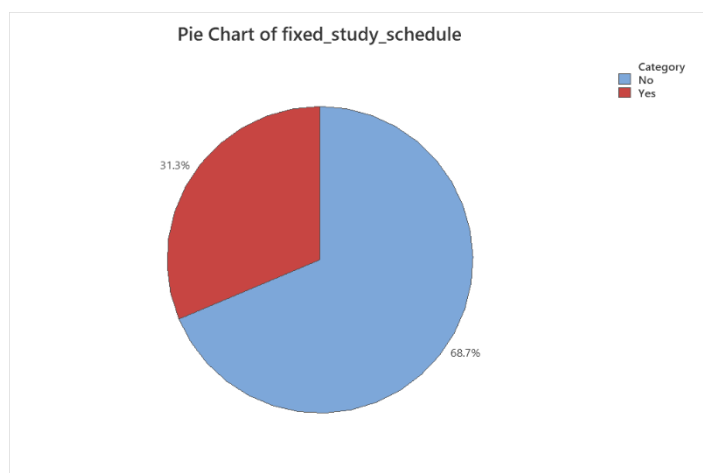


Figure 3 depicts that the highest percentage of students have a study schedule, which 68.7%, and only 31.3% of students do not have a study schedule.

4.1.4 Composition of the sample with respect to gender and fixed study schedule

Table 01 - Tabulated statistics of Gender, digital tools usage

Rows: Gender Columns: digital_tool_usage

	Daily	Never	Occasionally	Weekly	All
Female	12	3	19	4	38
	31.58	7.89	50.00	10.53	100.00
	54.55	75.00	47.50	23.53	45.78
	14.46	3.61	22.89	4.82	45.78
Male	10	1	21	13	45
	22.22	2.22	46.67	28.89	100.00
	45.45	25.00	52.50	76.47	54.22
	12.05	1.20	25.30	15.66	54.22
All	22	4	40	17	83
	26.51	4.82	48.19	20.48	100.00
	100.00	100.00	100.00	100.00	100.00
	26.51	4.82	48.19	20.48	100.00

Cell Contents
Count
% of Row
% of Column
% of Total

According to Table 01, females have the highest percentage for daily and never usage of digital tools, while males have the highest percentage for occasional and weekly usage.

4.1.5 Composition of the sample with respect to gender and digital tools usage

Table 02 - Tabulated statistics of Gender, Fixed study schedule

Rows: Gender Columns: fixed_study_schedule

	No	Yes	All
Female	25	13	38
	65.79	34.21	100.00
	43.86	50.00	45.78
	30.12	15.66	45.78
Male	32	13	45
	71.11	28.89	100.00
	56.14	50.00	54.22
	38.55	15.66	54.22
All	57	26	83
	68.67	31.33	100.00
	100.00	100.00	100.00
	68.67	31.33	100.00

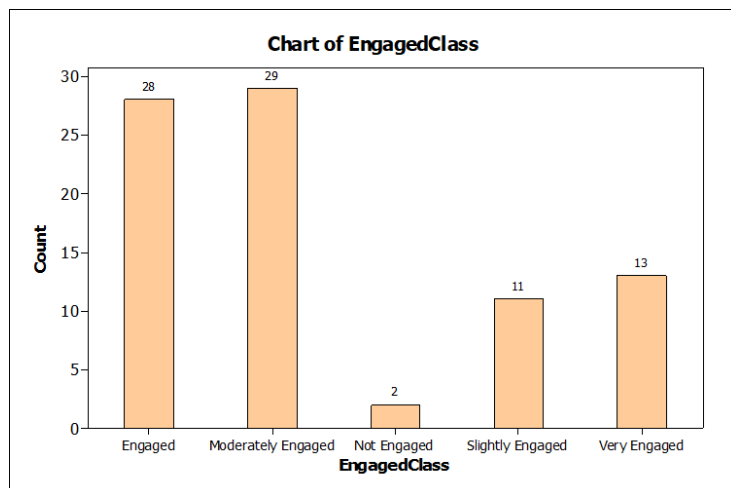
Cell Contents
Count
% of Row
% of Column
% of Total

Table 2 shows that males do not have a study schedule than that of females.

4.2. Frequency Distribution of Quantitative Independent Variable

4.2.1 Frequency Distribution of Engaged Class

Figure 04 – Bar chart of Engaged Class



According to Figure 04, engaged and moderately engaged have a higher number when compared to other classes. The not engaged class has the lowest number.

4.2.2 Frequency Distribution of Academic satisfaction

Figure 05 – Bar chart of Academic satisfaction

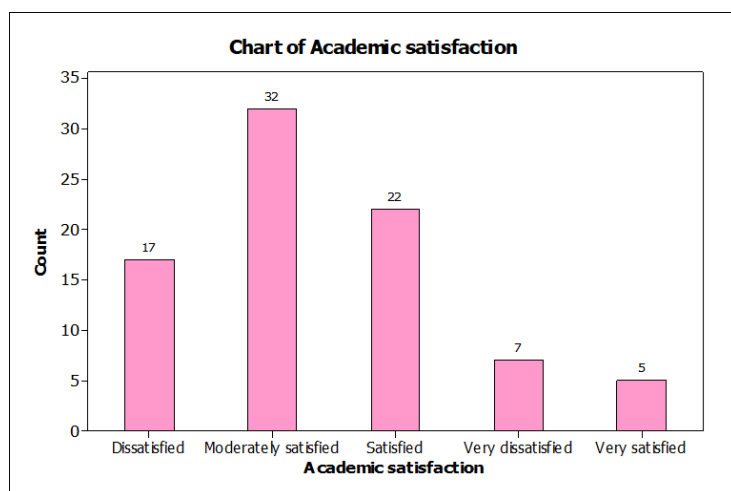


Figure 05 shows that most individuals are moderately satisfied with their academic performance, while only five are very satisfied.

4.3. Distribution of Quantitative Independent Variables

4.3.1 Distribution of Study Hours

Figure 06 -Boxplot of Study Hours

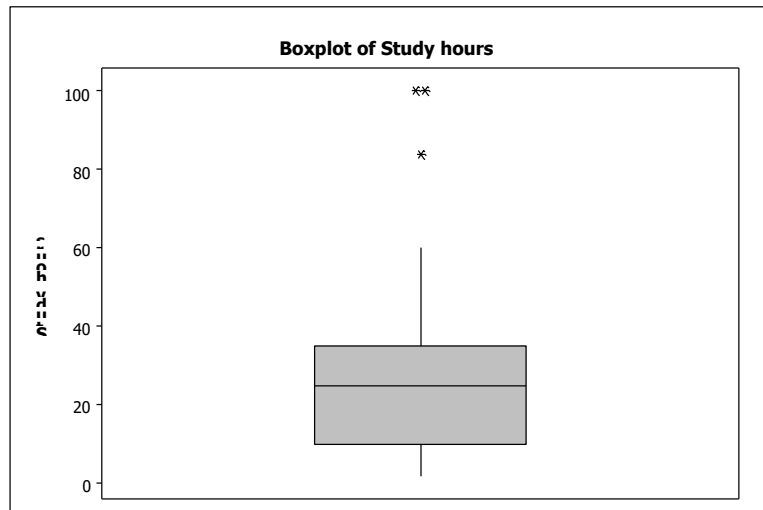


Figure 07 -Dot plot of Study Hours

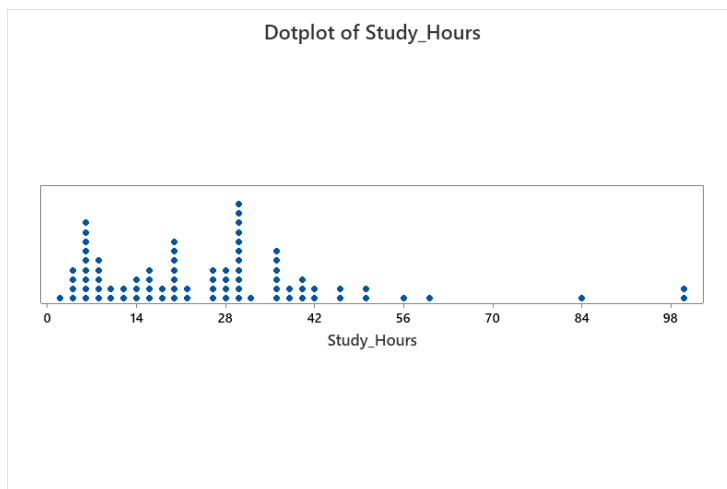


Table 03- Descriptive Statistics: Study Hours

Variable	Mean	SE Mean	StDev	Minimum	Q1	Median	Q3	Maximum
Study_Hours	25.59	2.12	19.35	2.00	10.00	25.00	35.00	100.00

According to Figure 07 and Table 03, study hours are not symmetrically distributed, but it has median value of 25 hours.

4.3.2 Distribution of attendance

Figure 08 -Boxplot of Attendance

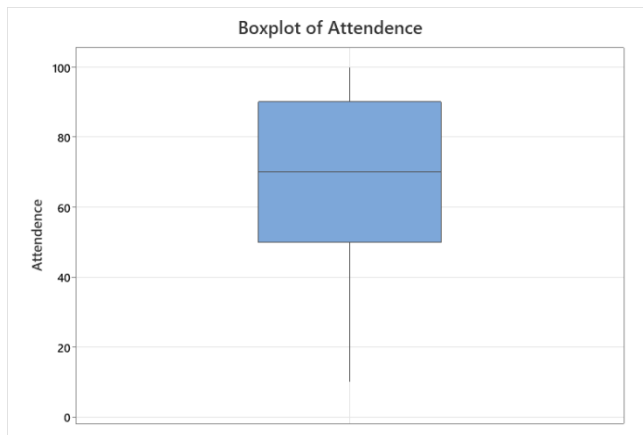


Figure 09 -Dot plot of Attendance

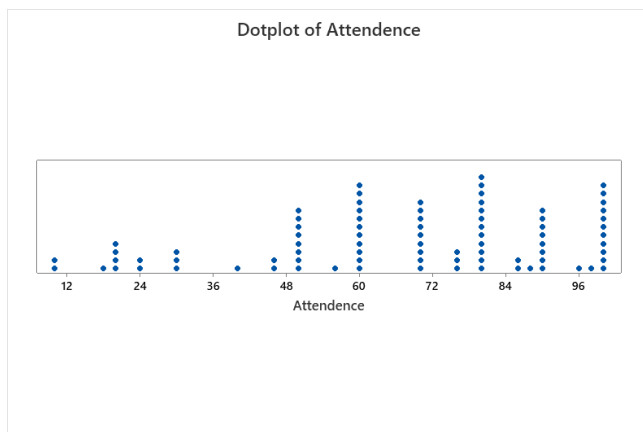


Table 04-Descriptive Statistics: Attendance

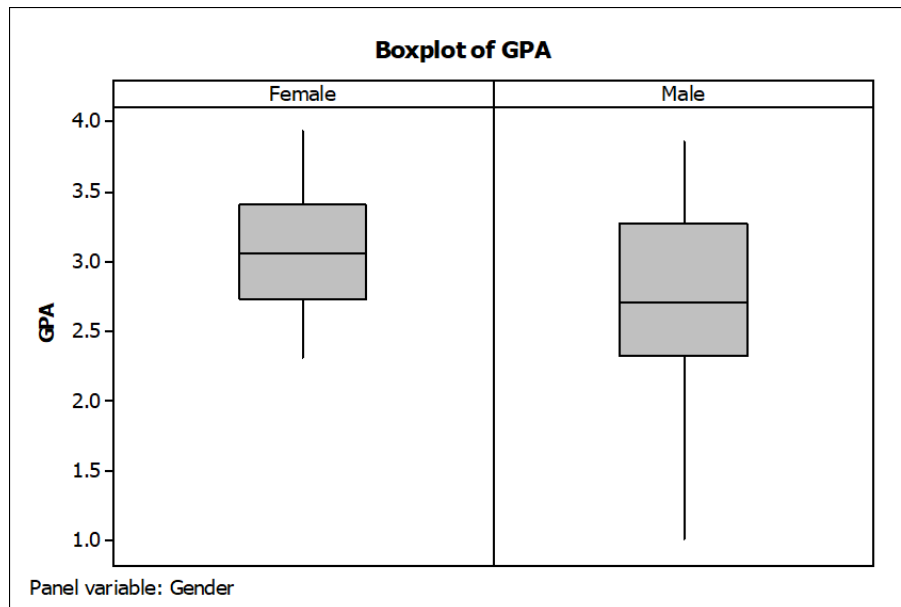
Variable	Mean	SE Mean	StDev	Minimum	Q1	Median	Q3	Maximum
Attendance	67.30	2.75	25.08	10.00	50.00	70.00	90.00	100.00

According to Figure 09 and Table 04, attendance is distributed with median value of 70, as a percentage.

4.4. Analysis of Dependent Variable

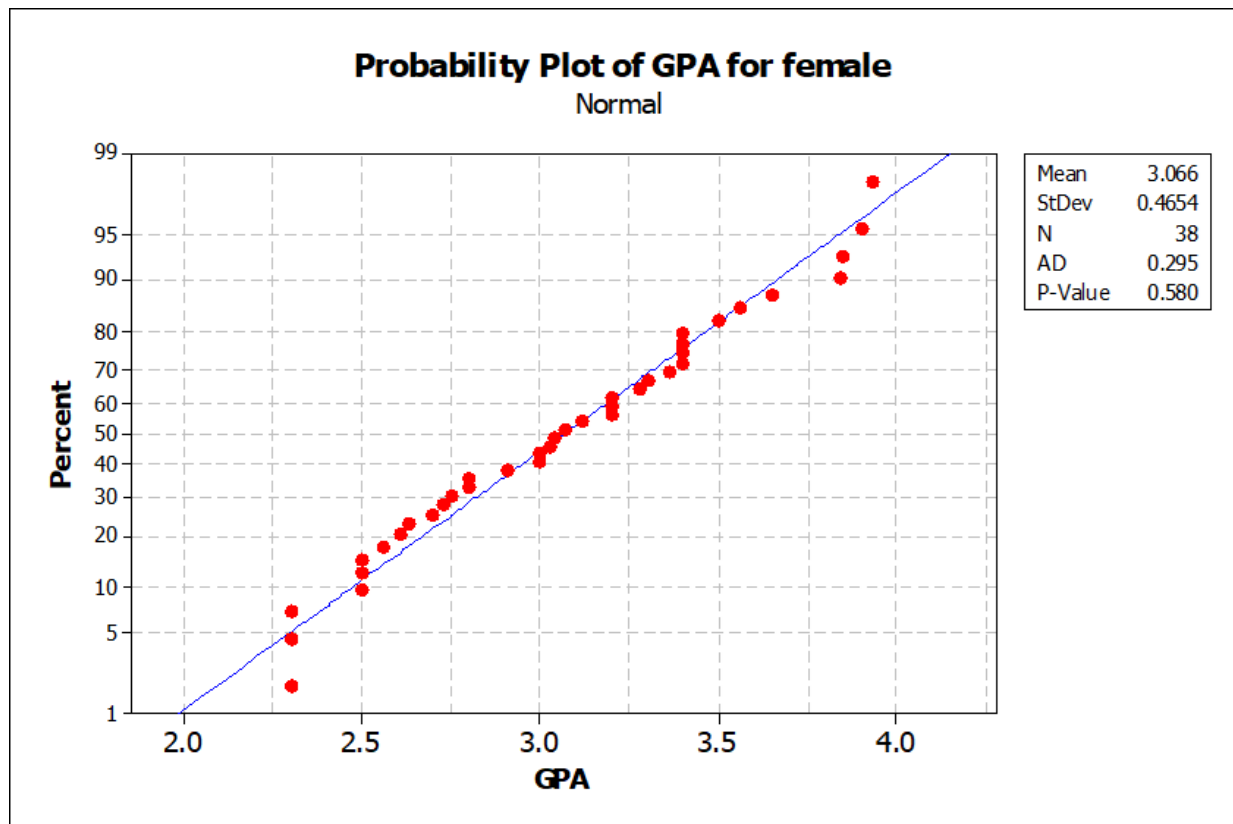
4.4.1 Association between Gender and GPA

Figure 10 – Box plot of GPA and Gender



According to Figure 10, it can be suggested that the median GPA seems to be different between Males and Females.

Figure 11 – Normality plot of GPA for females



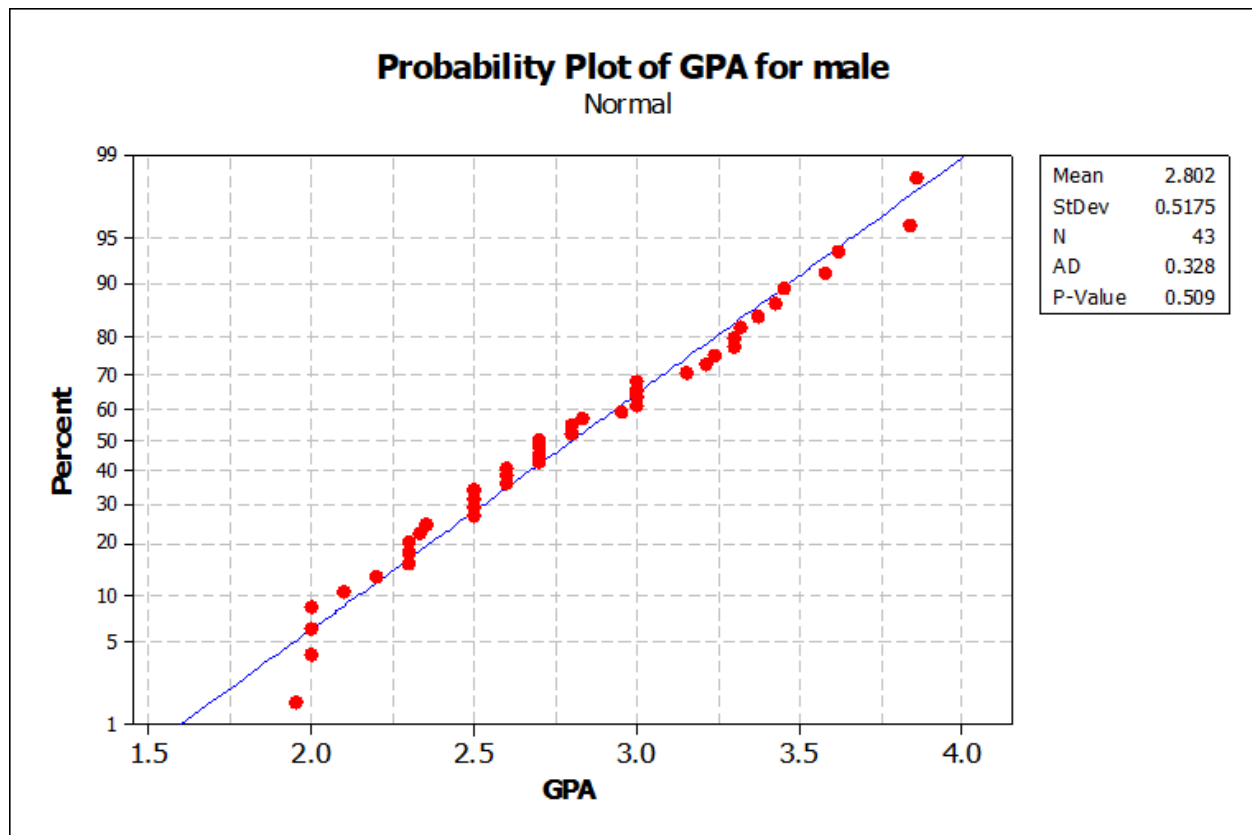
Hypothesis to be tested,

H_0 : GPA of females is normally distributed.

H_1 : GPA of females is not normally distributed.

Figure 11 shows that the p-value is 0.580, and it is greater than 0.05. Therefore, we do not have enough evidence to reject H_0 at 5% level of significance. This concludes that the GPA of females is normally distributed.

Figure 12 – Normality plot of GPA for male



Hypothesis to be tested,

H_0 : GPA of male are normally distributed.

H_1 : GPA of male are not normally distributed.

Figure 12 shows that the p-value is 0.509, and it is greater than 0.05. Therefore, we do not have enough evidence to reject H_0 at 5% level of significance. This concludes that GPA of males is normally distributed.

Therefore, we can use a parametric two sample t test, before that, we want to check whether variances are equal.

Table 05 - Test and CI for Two Variances of GPA vs Gender

Method

σ_1 : standard deviation of GPA when Gender = Female
 σ_2 : standard deviation of GPA when Gender = Male
 Ratio: σ_1/σ_2
 F method was used. This method is accurate for normal data only.

Descriptive Statistics

Gender	N	StDev	Variance	95% CI for σ
Female	38	0.455	0.207	(0.371, 0.589)
Male	45	0.598	0.357	(0.495, 0.755)

Ratio of Standard Deviations

Estimated Ratio	95% CI for Ratio using F
0.761796	(0.559, 1.047)

Test

Null hypothesis $H_0: \sigma_1 / \sigma_2 = 1$
 Alternative hypothesis $H_1: \sigma_1 / \sigma_2 \neq 1$
 Significance level $\alpha = 0.05$

Test				
Method	Statistic	DF1	DF2	P-Value
F	0.58	37	44	0.093

Hypothesis to be tested,

H_0 : variances of GPA of females and variance of GPA of males are equal.

H_1 : variances of GPA of females and variance of GPA of males are not equal.

According to Table 05, the p-value of the F test (0.093) is greater than 0.05. Therefore, we do not have enough evidence to reject H_0 at 5% level of significance. Then variances of GPA of males and females are equal.

Table 06 - Two-Sample T-Test and CI of GPA, Gender

Method μ_1 : population mean of GPA when Gender = Female μ_2 : population mean of GPA when Gender = MaleDifference: $\mu_1 - \mu_2$ *Equal variances are assumed for this analysis.***Descriptive Statistics: GPA**

Gender	N	Mean	StDev	SE Mean
Female	38	3.074	0.455	0.074
Male	45	2.778	0.598	0.089

Estimation for Difference

		95% CI for
Difference	Pooled StDev	Difference
0.297	0.537	(0.061, 0.532)

TestNull hypothesis $H_0: \mu_1 - \mu_2 = 0$ Alternative hypothesis $H_1: \mu_1 - \mu_2 \neq 0$

T-Value	DF	P-Value
2.50	81	0.014

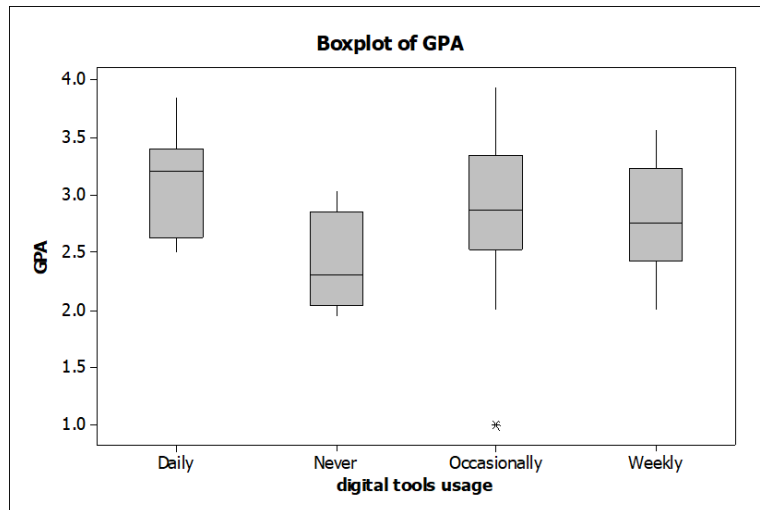
Hypotheses to be tested,

 H_0 : Mean GPA of females is equal to the mean GPA of males H_1 : Mean GPA of females is not equal to the mean GPA of males

According to Table 06, the p-value of the two-sample t-test (0.014) is less than 0.05, we have enough evidence to reject H_0 at 5% level of significance. It proves that there is an association between GPA and gender.

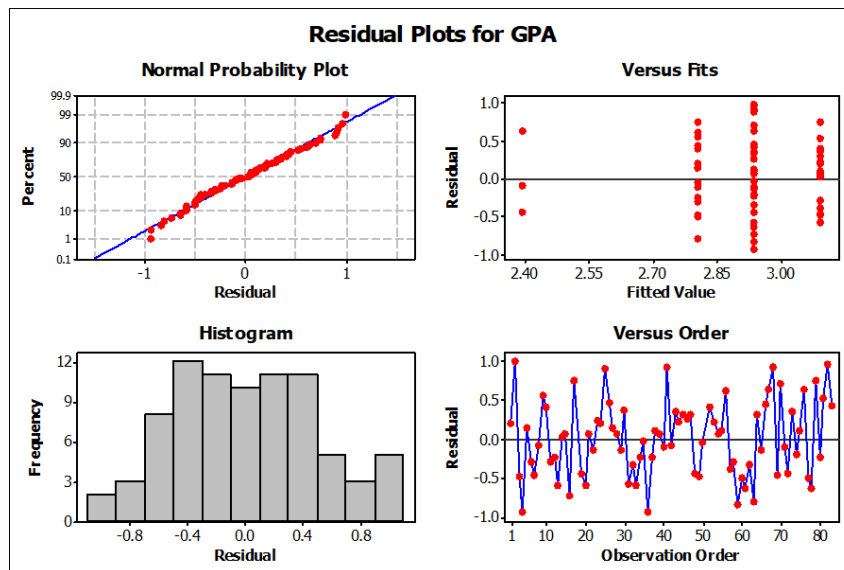
4.4.2 Association between GPA and Digital tool usage

Figure 13 – Box plot of GPA and Digital tool usage



According to Figure 13, there are outliers in GPA at each level of digital tool usage. The median GPA for digital tool usage seems different.

Figure 14 – Residual plots of GPA and Digital tool usage



According to Figure 14, it can be concluded that the constant variance assumption is not satisfied. Therefore, it can be concluded that the assumptions for one-way ANOVA are not satisfied.

Therefore, the Kruskal-Wallis test is applied.

Table 07 - Kruskal-Wallis Test of GPA versus digital tools usage

Descriptive Statistics

digital_tool_usage	N	Median	Mean Rank	Z-Value
Daily	22	3.20	49.8	1.78
Never	4	2.40	20.3	-1.85
Occasionally	40	2.87	42.1	0.05
Weekly	17	2.75	36.6	-1.03
Overall	83		42.0	

TestNull hypothesis H_0 : All medians are equalAlternative hypothesis H_1 : At least one median is different

Method	DF	H-Value	P-Value
Not adjusted for ties	3	6.42	0.093
Adjusted for ties	3	6.44	0.092

The chi-square approximation may not be accurate when some sample sizes are less than 5.

Hypothesis to be tested,

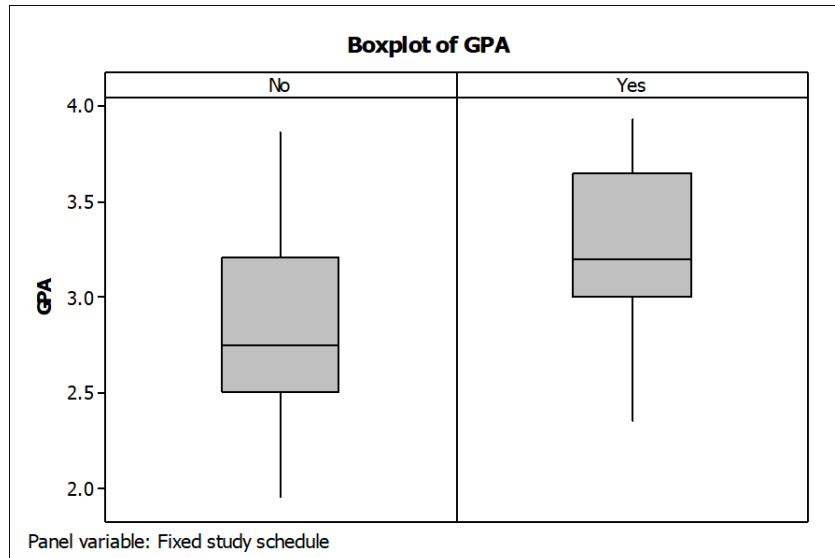
H_0 : Median GPA for individuals who use all types of digital tools is equal.

H_1 : Median GPA for individuals who use all types of digital tools is not equal.

Table 07 shows $P\text{-value} = 0.092 > 0.05$. Therefore, H_0 is not rejected at 5% level of significance. It concludes that the median GPAs are same for all types of digital tools.

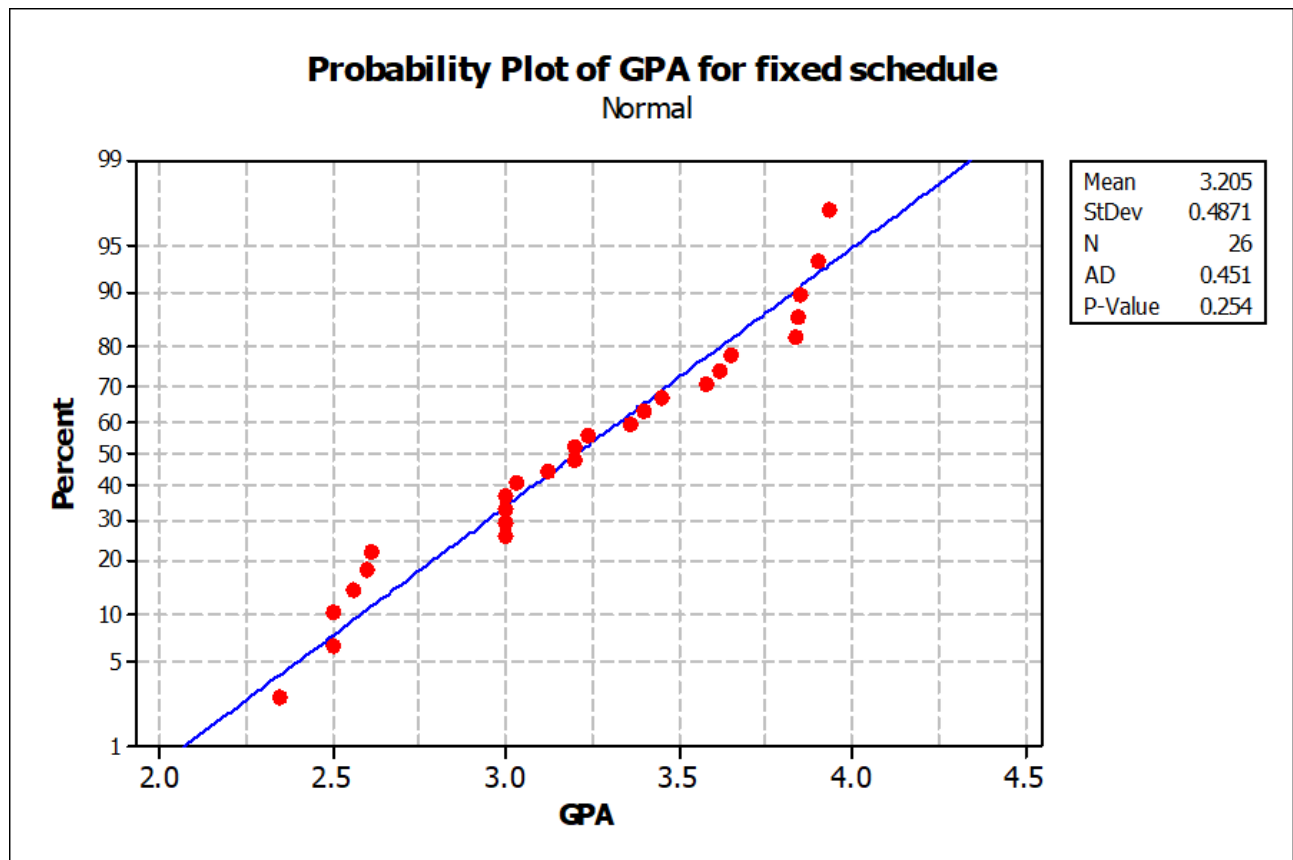
4.4.3 Association between GPA and Fixed time schedule

Figure 15 – Box plot of GPA and Fixed time schedule



According to Figure 15, the median GPA of the individuals who have a fixed study schedule is higher than that of those who do not.

Figure 16 – Normality plot of GPA for fixed schedule



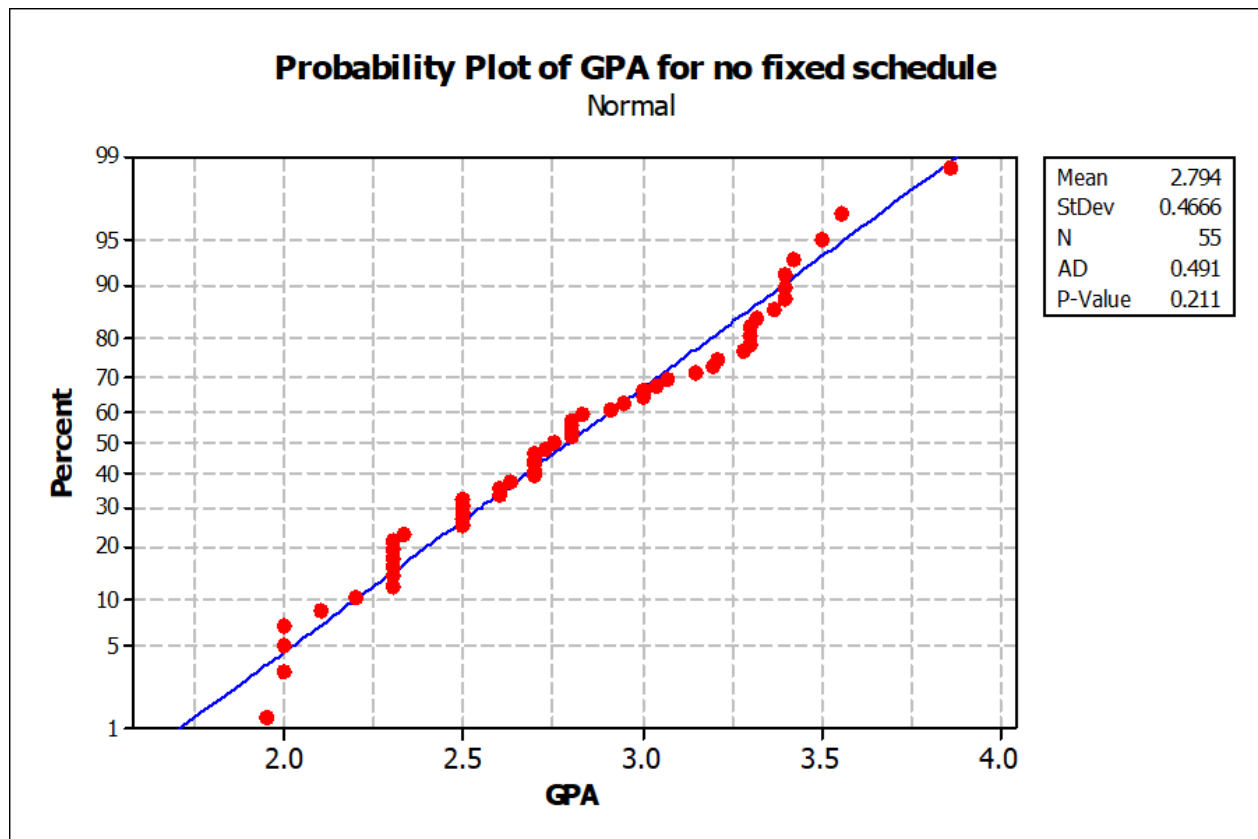
Hypothesis to be tested,

H_0 : GPA of individuals who have fixed schedules is normally distributed.

H_1 : GPA of individuals who have fixed schedules is not normally distributed.

Figure 16 shows that the p-value is 0.254, and it is greater than 0.05. Therefore, we do not have enough evidence to reject H_0 at 5% level of significance. This concludes that the GPA of individuals who have fixed schedules is normally distributed.

Figure 17 – Normality plot of GPA for no fixed schedule



Hypothesis to be tested,

H_0 : GPA of individuals who have no fixed schedules are normally distributed

H_1 : GPA of individuals who have no fixed schedules are not normally distributed

Figure17 shows that the p-value is 0.211 and it is greater than 0.05. Therefore, we do not have enough evidence to reject H_0 at 5% level of significance. It concludes that GPA of individuals who have no fixed schedules are normally distributed.

Table 08 - Test and CI for Two Variances of GPA vs Fixed study schedule

Method

σ_1 : standard deviation of GPA when fixed_study_schedule = No
 σ_2 : standard deviation of GPA when fixed_study_schedule = Yes
 Ratio: σ_1/σ_2
 F method was used. This method is accurate for normal data only.

Descriptive Statistics

fixed_study_schedule	N	StDev	Variance	95% CI for σ
No	57	0.539	0.291	(0.455, 0.662)
Yes	26	0.481	0.231	(0.377, 0.664)

Ratio of Standard Deviations

Estimated Ratio	95% CI for Ratio using F
1.12119	(0.781, 1.540)

Test

Null hypothesis $H_0: \sigma_1 / \sigma_2 = 1$
 Alternative hypothesis $H_1: \sigma_1 / \sigma_2 \neq 1$
 Significance level $\alpha = 0.05$

Test				
Method	Statistic	DF1	DF2	P-Value
F	1.26	56	25	0.539

Hypothesis to be tested,

H_0 : the variance of GPA of individuals who have fixed schedules and the variance of GPA of individuals who have no fixed schedules are equal.

H_1 : the variance of GPA of individuals who have fixed schedules and the variance of GPA of individuals who have no fixed schedules are not equal.

According to Table 08, the p-value of the F test (0.539) is greater than 0.05. Therefore, we do not have enough evidence to reject H_0 at 5% level of significance. Then the variances of GPA of individuals who have fixed schedules and the variance of GPA of individuals who have no fixed schedules are equal.

Table 09 - Two-Sample T-Test and CI of GPA, Fixed study schedule

Method

μ_1 : population mean of GPA when fixed_study_schedule = No

μ_2 : population mean of GPA when fixed_study_schedule = Yes

Difference: $\mu_1 - \mu_2$

Equal variances are assumed for this analysis.

Descriptive Statistics: GPA

fixed_study_schedule	N	Mean	StDev	SE Mean
No	57	2.782	0.539	0.071
Yes	26	3.201	0.481	0.094

Estimation for Difference

		95% CI for	
Difference	Pooled StDev	Difference	
-0.418	0.522	(-0.664, -0.173)	

Test

Null hypothesis $H_0: \mu_1 - \mu_2 = 0$

Alternative hypothesis $H_1: \mu_1 - \mu_2 \neq 0$

T-Value	DF	P-Value
-3.39	81	0.001

Hypotheses to be tested,

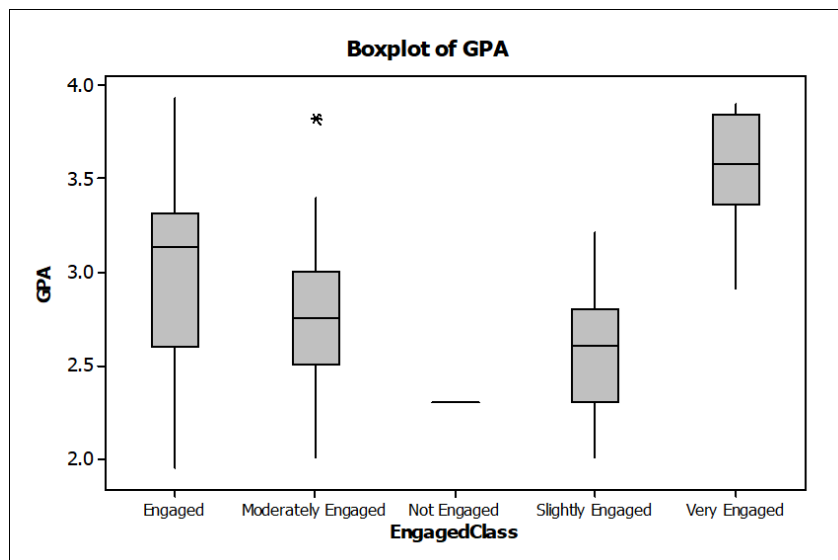
H_0 : Mean GPA of females is equal to the mean GPA of males

H_1 : Mean GPA of females is not equal to the mean GPA of males

According to Table 09, the p-value of the two-sample t-test (0.001) is less than 0.05, we have enough evidence to reject H_0 at 5% level of significance. It proves that there is an association between GPA and fixed study schedule.

4.4.4 Association between GPA and engaged in class

Figure 18 – Box plot of GPA and engaged in class



According to Figure 18, there are outliers in GPA according to engagement in class. The median GPA according to engagement in class seems different.

Therefore, the Kruskal-Wallis test is applied.

It is ordinal state therefore, we are going to use non-parametric tests.

Table 10 - Kruskal-Wallis Test of GPA versus EngagedClass

Descriptive Statistics

engaged_class	N	Median	Mean Rank	Z-Value
1	2	2.300	10.5	-1.87
2	11	2.600	23.9	-2.67
3	29	2.750	34.0	-2.21
4	28	3.135	46.3	1.16
5	13	3.580	70.7	4.67
Overall	83		42.0	

Test

Null hypothesis H_0 : All medians are equal

Alternative hypothesis H_1 : At least one median is different

Method	DF	H-Value	P-Value
Not adjusted for ties	4	32.11	0.000
Adjusted for ties	4	32.18	0.000

The chi-square approximation may not be accurate when some sample sizes are less than 5.

Hypothesis to be tested,

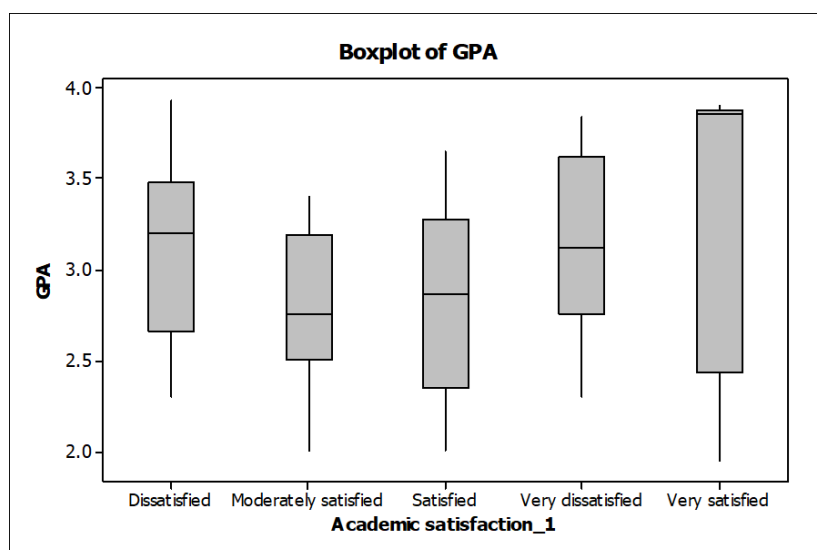
H0: Median GPA for individuals according to engagement in class is equal.

H1: Median GPA for individuals according to engagement in class is not equal.

Table 10 shows $P\text{-value} = 0.0001 < 0.05$, H0 is rejected at 5% level of significance. We have enough evidence to conclude that the median GPAs are not the same for all types of digital tools.

4.4.5 Association between GPA and academic satisfaction

Figure 19 – Box plot of GPA and academic satisfaction



According to Figure 19. There are no outliers in GPA according to academic satisfaction. The median GPA according to academic satisfaction seems different.

It is an ordinal scale; therefore, we are going to use non-parametric tests.

Therefore, the Kruskal-Wallis test is applied.

Table 11 - Kruskal-Wallis Test of GPA versus academic satisfaction

Descriptive Statistics

Academic_satisfaction	N	Median	Mean Rank	Z-Value
1	5	3.850	57.2	1.45
2	22	2.865	37.8	-0.96
3	32	2.765	36.3	-1.72
4	17	3.200	49.3	1.40
5	7	3.120	53.0	1.26
Overall	83		42.0	

Test

Null hypothesis H_0 : All medians are equal
 Alternative hypothesis H_1 : At least one median is different

Method	DF	H-Value	P-Value
Not adjusted for ties	4	7.52	0.111
Adjusted for ties	4	7.54	0.110

Hypothesis to be tested,

H_0 : Median GPA for individuals according to academic satisfaction is equal.

H_1 : Median GPA for individuals according to academic satisfaction is not equal.

Table 11 shows $P\text{-value} = 0.110 > 0.05$., H_0 is not rejected at 5% level of significance. We do not have enough evidence to reject H_0 . It is concluded that median GPAs are same for all types of academic satisfaction.

4.4.6 Association between GPA and study hours

Figure 20 – Scatter plot of GPA and study hours

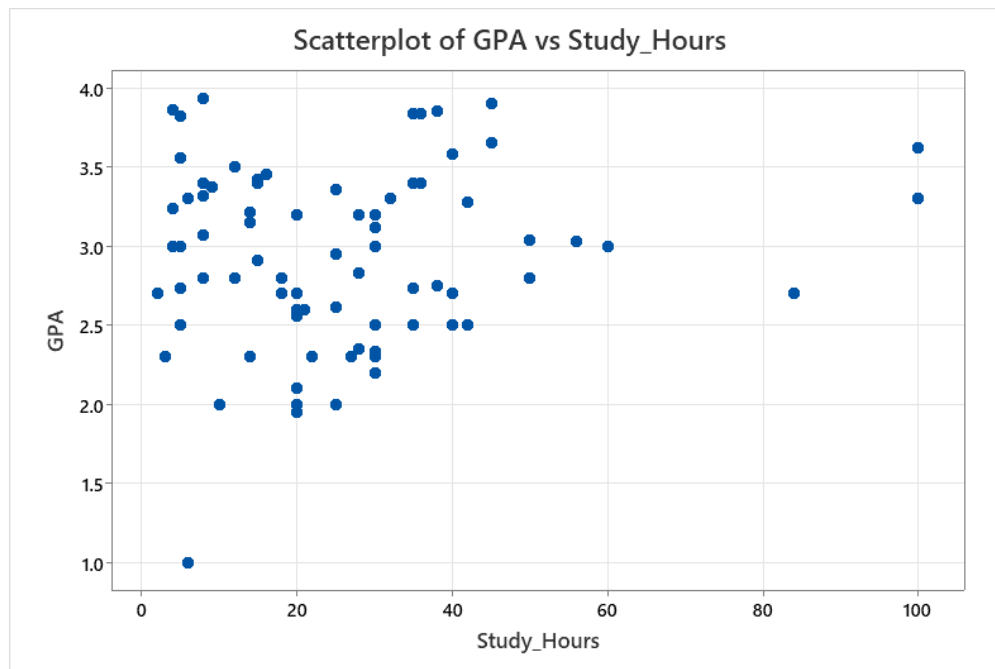


Table 12 – Correlation between GPA and study hours

Pearson correlation of GPA and Study hours	0.138
P-Value	0.214

H_0 : There is no significant correlation between GPA and study hours. ($\rho = 0$)

H_1 : There is a significant correlation between GPA and study hours. ($\rho \neq 0$)

According to Table 12, P-value = 0.214, this p-value is much greater than 0.05, suggesting that the correlation is not statistically significant. This means there is no evidence of a meaningful association between GPA and study hours based on this data.

4.4.7 Association between GPA and attendance

Figure 21 – Scatter plot of GPA and attendance

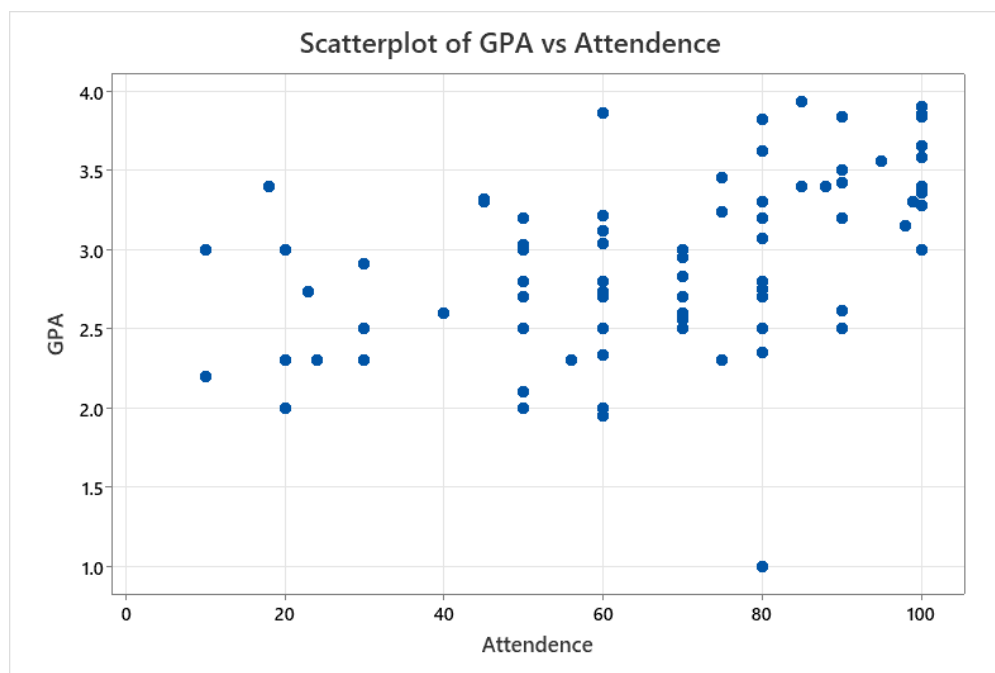


Table 13 – Correlation between GPA and Attendance

Pearson correlation of GPA and Attendance	0.441
P-Value	0.000

H_0 : There is no significant correlation between GPA and attendance ($\rho = 0$)

H_1 : There is a significant correlation between GPA and attendance. ($\rho \neq 0$)

According to Table 13, the p-value = 0.000, this value is statistically significant ($p < 0.05$). It indicates strong evidence that the observed relationship between GPA and attendance is not due to chance.

Figure 21 and Table 13 depict a Pearson correlation coefficient = 0.441, this indicates a moderate positive correlation between GPA and attendance. Higher attendance is associated with higher GPA values.

4.5. Fitting the Regression Model

4.5.1 Effect of qualitative predictors on GPA

Table 14 - General Regression Analysis of GPA versus Gender, digital tool, and fixed study schedule

Coefficients

Term	Coef	SE Coef	95% CI	T-Value	P-Value	VIF
Constant	3.118	0.158	(2.804, 3.432)	19.78	0.000	
Gender						
Female	0.040	0.270	(-0.498, 0.578)	0.15	0.883	6.74
Male	0.000000	0.000000	(0.000000, 0.000000)	*	*	*
digital_tool_usage						
Daily	0.325	0.199	(-0.072, 0.721)	1.63	0.107	2.87
Never	-0.747	0.491	(-1.725, 0.231)	-1.52	0.132	4.11
Occasionally	-0.207	0.167	(-0.540, 0.126)	-1.24	0.220	2.59
Weekly	0.000000	0.000000	(0.000000, 0.000000)	*	*	*
fixed_study_schedule						
No	-0.421	0.114	(-0.648, -0.194)	-3.69	0.000	1.04
Yes	0.000000	0.000000	(0.000000, 0.000000)	*	*	*
Gender*digital_tool_usage						
Female Daily	-0.242	0.338	(-0.916, 0.432)	-0.72	0.476	5.26
Female Never	0.480	0.610	(-0.735, 1.695)	0.79	0.434	4.82
Female Occasionally	0.583	0.309	(-0.032, 1.199)	1.89	0.063	6.27
Female Weekly	0.000000	0.000000	(0.000000, 0.000000)	*	*	*
Male Daily	0.000000	0.000000	(0.000000, 0.000000)	*	*	*
Male Never	0.000000	0.000000	(0.000000, 0.000000)	*	*	*
Male Occasionally	0.000000	0.000000	(0.000000, 0.000000)	*	*	*
Male Weekly	0.000000	0.000000	(0.000000, 0.000000)	*	*	*

Model Summary

S	R-sq	R-sq(adj)	PRESS	R-sq(pred)	AICc	BIC
0.472411	34.45%	27.36%	*	*	124.59	145.72

A general regression analysis was conducted to examine the influence of qualitative factors (gender, digital tools usage, and fixed study schedules) on students' Grade Point Average (GPA).

For gender, we use one dummy variable (d1).

d1=0 for male

d1=1 for female

Model

$$Y = \beta_0 + \beta_1(d1)$$

male (d1=0): $Y = \beta_0 + \beta_1(0)$

female (d1=1): $Y = \beta_0 + \beta_1(1)$

Hypothesis to be tested,

H0: $\beta_1 = 0$ vs H1: $\beta_1 \neq 0$

Since p-value $0.883 > 0.05$ we do not reject H0. Gender is not significant.

For fixed study schedule, we used one dummy variable(d_1) $d_1=0$ for fixed study schedule
 $d_1=1$ for no fixed study schedule

Model : $Y = \beta_0 + \beta_2(d_1)$

Yes ($d_1=0$) : $Y = \beta_0 + \beta_2(0)$

No ($d_1=1$) : $Y = \beta_0 + \beta_2(1)$

Hypothesis to be tested,

$H_0: \beta_2 = 0$ vs $H_1: \beta_2 \neq 0$

Since $p\text{-value} (0.000) < \alpha(0.05)$ we reject H_0 . Therefore, the fixed study schedule is significant.

Since the digital tool usage has 4 levels, we use three dummy variables (d_2, d_3, d_4)

Weekly : $d_2, d_3, d_4 = 0$

Daily : $d_2 = 1, d_3=0, d_4=0$

Never : $d_2 = 0, d_3 = 1, d_4=0$

occasionally : $d_2=0, d_3=0, d_4=1$

Model : $Y = \beta_0 + \beta_2(d_1) + \beta_3(d_2) + \beta_4(d_3) + \beta_5(d_4)$

Weekly : $Y = \beta_0 + \beta_2(d_1)$

Daily : $Y = \beta_0 + \beta_2(d_1) + \beta_3(d_2)$

Never : $Y = \beta_0 + \beta_2(d_1) + \beta_4(d_3)$

Occasionally : $Y = \beta_0 + \beta_2(d_1) + \beta_5(d_5)$

At a significance level of $\alpha = 0.05$, none of the main effect categories for digital tool usage Daily ($p = 0.107$), Never ($p = 0.132$), or Occasionally ($p = 0.220$) were statistically significant relative to the reference level (Weekly).

Furthermore, none of the interaction terms between Gender and Digital Tool Usage reached statistical significance ($p > 0.05$).

The only statistically significant predictor in this model is Fixed Study Schedule (No) with a coefficient of $\beta = -0.421$ ($p < 0.001$), indicating that students without a fixed study schedule have a significantly lower GPA compared to those with a fixed schedule, holding all other variables constant.

Therefore, the model (only with qualitative predictors) is,

$$\begin{aligned} \text{GPA} = & 3.118 + 0.040(\text{Gender}_{\text{Female}}) + 0.325(\text{DigitalTool}_{\text{Daily}}) - 0.747(\text{DigitalTool}_{\text{Never}}) \\ & - 0.207(\text{DigitalTool}_{\text{Occasionally}}) - 0.421(\text{FixedStudySchedule}_{\text{No}}) \\ & - 0.242(\text{Gender}_{\text{Female}} \times \text{DigitalTool}_{\text{Daily}}) \\ & + 0.480(\text{Gender}_{\text{Female}} \times \text{DigitalTool}_{\text{Never}}) \\ & + 0.583(\text{Gender}_{\text{Female}} \times \text{DigitalTool}_{\text{Occasionally}}) \end{aligned}$$

4.5.2 Effect of quantitative predictors on GPA

Figure 22 – Matrix plot of GPA and quantitative predictors

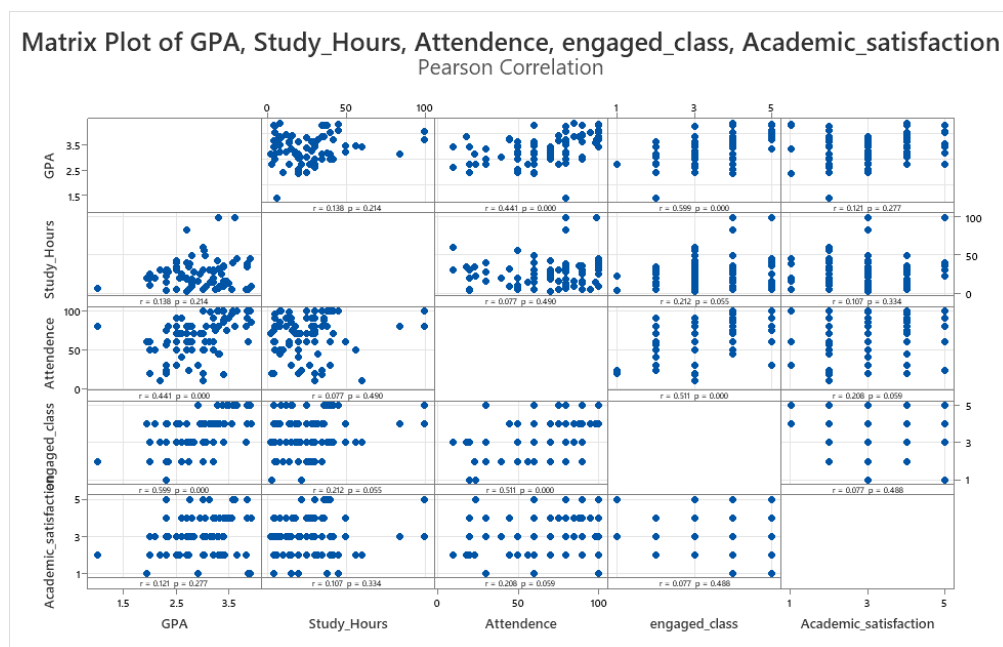


Table 15 – Correlations of GPA, Study hours, Attendance, EngagedClass, Academic satisfaction

Pairwise Pearson Correlations

Sample 1	Sample 2	N	Correlation	95% CI for ρ	P-Value
Study Hours	GPA	83	0.138	(-0.080, 0.343)	0.214
Attendance	GPA	83	0.441	(0.249, 0.600)	0.000
engaged_class	GPA	83	0.599	(0.440, 0.721)	0.000
Academic_satisfaction	GPA	83	0.121	(-0.098, 0.328)	0.277
Attendance	Study_Hours	83	0.077	(-0.141, 0.288)	0.490
engaged_class	Study_Hours	83	0.212	(-0.004, 0.409)	0.055
Academic_satisfaction	Study_Hours	83	0.107	(-0.111, 0.316)	0.334
engaged_class	Attendance	83	0.511	(0.332, 0.655)	0.000
Academic_satisfaction	Attendance	83	0.208	(-0.008, 0.406)	0.059
Academic_satisfaction	engaged_class	83	0.077	(-0.141, 0.288)	0.488

According to Table 15, GPA is most strongly correlated with class engagement ($r = 0.599$, $p < 0.001$) and lecture attendance ($r = 0.441$, $p < 0.001$). Study hours and academic satisfaction show weak, statistically non-significant correlations with GPA ($p > 0.05$).

4.5.3 Best-fitted regression model

Table 16 -General Regression Analysis: GPA versus EngagedClass, Gender, digital tool usage, fixed study schedule, attendance

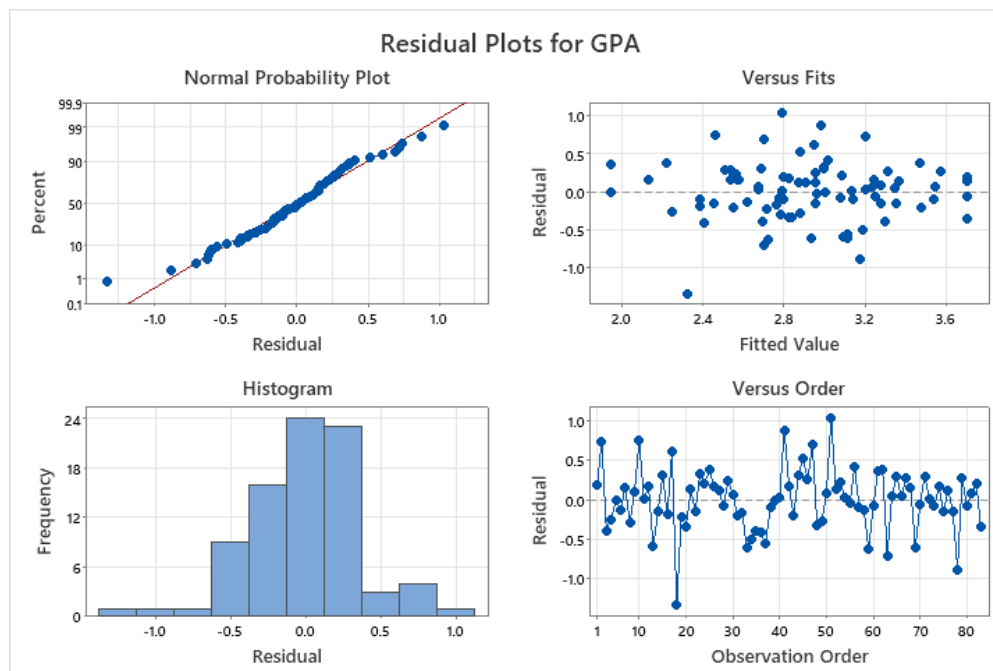
Regression Equation

Gender	digital_tool_usage	fixed_study_schedule	
Female	Daily	No	GPA = 1.950 + 0.2367 engaged_class + 0.00258 Attendance
Female	Daily	Yes	GPA = 2.175 + 0.2367 engaged_class + 0.00258 Attendance
Female	Never	No	GPA = 1.845 + 0.2367 engaged_class + 0.00258 Attendance
Female	Never	Yes	GPA = 2.071 + 0.2367 engaged_class + 0.00258 Attendance
Female	Occasionally	No	GPA = 2.038 + 0.2367 engaged_class + 0.00258 Attendance
Female	Occasionally	Yes	GPA = 2.263 + 0.2367 engaged_class + 0.00258 Attendance
Female	Weekly	No	GPA = 1.761 + 0.2367 engaged_class + 0.00258 Attendance
Female	Weekly	Yes	GPA = 1.986 + 0.2367 engaged_class + 0.00258 Attendance
Male	Daily	No	GPA = 1.934 + 0.2367 engaged_class + 0.00258 Attendance
Male	Daily	Yes	GPA = 2.159 + 0.2367 engaged_class + 0.00258 Attendance
Male	Never	No	GPA = 0.848 + 0.2367 engaged_class + 0.00258 Attendance
Male	Never	Yes	GPA = 1.074 + 0.2367 engaged_class + 0.00258 Attendance
Male	Occasionally	No	GPA = 1.649 + 0.2367 engaged_class + 0.00258 Attendance
Male	Occasionally	Yes	GPA = 1.874 + 0.2367 engaged_class + 0.00258 Attendance
Male	Weekly	No	GPA = 1.836 + 0.2367 engaged_class + 0.00258 Attendance
Male	Weekly	Yes	GPA = 2.062 + 0.2367 engaged_class + 0.00258 Attendance

Coefficients

Term	Coef	SE Coef	95% CI	T-Value	P-Value	VIF
Constant	2.062	0.249	(1.565, 2.559)	8.27	0.000	
engaged_class	0.2367	0.0598	(0.1174, 0.3560)	3.95	0.000	1.71
Attendance	0.00258	0.00221	(-0.00183, 0.00700)	1.17	0.248	1.50
Gender						
Female	-0.076	0.239	(-0.552, 0.401)	-0.32	0.752	6.96
Male	0.000000	0.000000	(0.000000, 0.000000)	*	*	*
digital_tool_usage						
Daily	0.097	0.179	(-0.259, 0.454)	0.54	0.588	3.06
Never	-0.988	0.430	(-1.845, -0.130)	-2.30	0.025	4.17
Occasionally	-0.188	0.147	(-0.481, 0.105)	-1.28	0.206	2.65
Weekly	0.000000	0.000000	(0.000000, 0.000000)	*	*	*
fixed_study_schedule						
No	-0.226	0.106	(-0.438, -0.014)	-2.12	0.037	1.20
Yes	0.000000	0.000000	(0.000000, 0.000000)	*	*	*
Gender*digital_tool_usage						
Female Daily	0.092	0.301	(-0.509, 0.693)	0.30	0.762	5.52
Female Never	1.072	0.543	(-0.011, 2.155)	1.97	0.052	5.05
Female Occasionally	0.465	0.271	(-0.076, 1.006)	1.71	0.091	6.38
Female Weekly	0.000000	0.000000	(0.000000, 0.000000)	*	*	*
Male Daily	0.000000	0.000000	(0.000000, 0.000000)	*	*	*
Male Never	0.000000	0.000000	(0.000000, 0.000000)	*	*	*
Male Occasionally	0.000000	0.000000	(0.000000, 0.000000)	*	*	*
Male Weekly	0.000000	0.000000	(0.000000, 0.000000)	*	*	*

Figure 23- Residual Plots for GPA



Regression Equation:

$$\begin{aligned} \text{GPA} = & 2.062 + 0.2367 \text{ engaged_class} + 0.00258 \text{ Attendance} - 0.076 \text{ Gender}_{\text{Female}} + 0.097 \\ & \text{digital_tool_usage}_{\text{Daily}} - 0.988 \text{ digital_tool_usage}_{\text{Never}} - 0.188 \text{ digital_tool_usage}_{\text{Occasionally}} - 0.226 \\ & \text{fixed_study_schedule}_{\text{No}} + 0.092 \text{ Gender} \times \text{digital_tool_usage}_{\text{Female Daily}} + 1.072 \\ & \text{Gender} \times \text{digital_tool_usage}_{\text{Female Never}} + 0.465 \text{ Gender} \times \text{digital_tool_usage}_{\text{Female Occasionally}} \end{aligned}$$

R-Sq = 51.72% Adjusted R-Sq = 45.01%

Variables and Interpretations:

1. Intercept (2.062, $p < 0.001$):

Represents the predicted GPA for a male student who uses digital tools weekly, follows a fixed study schedule (Yes), and has zero attendance and zero engagement score (engaged_class = 0). While zero attendance and engagement are unrealistic in practice, the intercept serves as the mathematical baseline for the model.

2. Engaged Class (Coef = 0.2367, $p < 0.001$):

For every one-unit increase in class engagement score (e.g., moving from “Moderately Engaged” to “Very Engaged”), GPA increases by 0.2367 points on average, holding all other variables constant. This is a highly statistically significant predictor in the model.

3. Attendance (Coef = 0.00258, $p = 0.248$):

For every 1% increase in attendance, GPA increases by 0.00258 points on average, holding all other variables constant. However, this effect is not statistically significant ($p > 0.05$).

4. Gender: Female (Coef = -0.076, $p = 0.752$):

Holding all other factors constant, female students who use digital tools weekly have an average GPA that is 0.076 points lower than male students who use digital tools weekly. This difference is not statistically significant ($p > 0.05$).

5. Digital Tool Usage (Reference: Weekly, Male):

- Daily (Coef = 0.097, $p = 0.588$): Male students who use digital tools daily score 0.097 points higher than male students who use them weekly (not statistically significant).

- Never (Coef = -0.988, $p = 0.025$): Male students who never use digital tools score 0.988 points lower than male students who use them weekly. This effect is statistically significant ($p < 0.05$).
- Occasionally (Coef = -0.188, $p = 0.206$): Male students who use tools occasionally score 0.188 points lower than male weekly users (not statistically significant).

6. Fixed Study Schedule: No (Coef = -0.226, $p = 0.037$):

Students who do not follow a fixed study schedule (No) have an average GPA that is 0.226 points lower than those who do follow a fixed schedule (Yes), holding all other variables constant. This effect is statistically significant ($p < 0.05$).

7. Gender×Digital Tool Usage Interaction:

- Female×Daily (Coef = 0.092, $p = 0.762$): The difference between daily and weekly tool usage is 0.092 points higher for females than for males (not statistically significant).
- Female×Never (Coef = 1.072, $p = 0.052$): The negative impact of never using digital tools is 1.072 points weaker for female students compared to male students (marginally significant at $p \approx 0.052$).
- Female×Occasionally (Coef = 0.465, $p = 0.091$): The difference between occasional and weekly tool usage is 0.465 points higher for females than for males (not statistically significant at $\alpha = 0.05$).

The best-fitting model shows that GPA is significantly driven by class engagement and having a fixed study schedule, with class engagement exerting the most substantial positive effect ($\beta = 0.2367$, $p < 0.001$). Following a fixed study schedule provides a statistically significant boost to GPA compared to not having one ($\beta = -0.226$, $p = 0.037$).

Among male students, completely avoiding digital tools is associated with a severe, statistically significant drop in GPA relative to weekly users (Coef = -0.988, $p = 0.025$). However, this negative penalty of never using digital tools is mitigated for female students, as captured by the marginally significant interaction term (Coef = 1.072, $p = 0.052$).

5. Discussion

This study aimed to investigate the relationship between academic performance and several potential predictors, including study hours, lecture and tutorial attendance, class engagement, use of digital and AI-powered tools, number of digital platforms used, and demographic factors such as gender and year of study. The primary objective was to identify which of these variables have a statistically significant effect on students' academic outcomes, as measured by self-reported GPA.

Initial descriptive statistics revealed a fairly balanced sample in terms of gender, with a slight predominance of male students. A fixed study schedule was maintained by 68.7% of students, whereas digital tool usage varied widely, with nearly half of the sample (48.2%) reporting occasional use. The average weekly study time was 25 hours, and average class attendance was 70.0%, indicating a generally moderate level of academic engagement.

Bivariate analyses revealed a range of associations between the predictor variables and academic performance. Gender appeared to play a role, with female students generally achieving higher academic performance than their male counterparts. Similarly, students who maintained a fixed study schedule tended to perform better academically than those who did not. Strong classroom engagement and consistent attendance were also positively linked to higher academic achievement. In contrast, time spent studying each week did not show a meaningful association with academic performance, suggesting that study quality may be more important than quantity.

The use of digital tools also appeared to influence academic outcomes. Students who regularly used digital learning resources tended to achieve better academic performance compared to those who rarely or never used them. While this association approached statistical significance, it indicates a potential benefit of consistent and purposeful use of educational technology.

A multiple regression model was constructed to identify the combined effect of qualitative and quantitative predictors on GPA, achieving an adjusted R^2 value of 45.01%. Among the predictors, class engagement (Coef = 0.2367, $p < 0.001$) emerged as a highly significant positive predictor of GPA. Conversely, never using digital tools (Coef = -0.988, $p = 0.025$) and lacking a fixed study schedule (Coef = -0.226, $p = 0.037$) had statistically significant negative impacts relative to their baseline groups (Weekly usage and having a fixed schedule). While main effects for gender and moderate digital tool frequencies were not statistically significant, the interaction term Female×Never (Coef = 1.072, $p = 0.052$) was marginally significant, indicating that the severe academic penalty of completely avoiding digital tools is less pronounced for female students than for male students.

Despite these findings, the study has some limitations. The use of self-reported GPA may introduce bias, and the cross-sectional nature of the study limits causal inference. The relatively small sample size ($n = 83$) and multicollinearity between attendance and engagement in class may explain the model's low R-squared value. Additionally, important factors such as mental health, socio-economic status, and specific learning strategies were not accounted for, potentially leaving out influential confounders. Nevertheless, the results highlight the value of structured learning practices, active classroom engagement, and thoughtful integration of digital tools in enhancing academic performance. These insights can be useful for educators aiming to design more effective teaching environments and for students striving to improve their academic success.

6. Conclusion

This study identified key predictors of academic performance among university students, emphasizing the role of behavioral and technological factors. The final regression model explained 51.72% of the variance in GPA (adjusted $R^2 = 45.01\%$).

Class engagement (Coef = 0.2367, $p < 0.001$) and maintaining a fixed study schedule (Coef = -0.226, $p = 0.037$) emerged as significant drivers of academic achievement. Conversely, completely avoiding digital tools produced a significant negative impact on performance (Coef = -0.988, $p = 0.025$), though this penalty was noticeably mitigated for female students, as evidenced by the marginally significant interaction term (Coef = 1.072, $p = 0.052$). Notably, lecture attendance ($p = 0.248$) and total study hours ($p = 0.214$) showed no statistically significant direct relationships with GPA in the model, suggesting that active engagement and study consistency outweigh sheer time duration.

These findings highlight the importance of active learning behaviors and consistent academic habits. Future studies could enhance model accuracy by incorporating additional factors such as mental health, socio-economic status, and learning strategies.

Although the model offers strong explanatory power, nearly half of the variability in GPA remains unexplained, pointing to the potential influence of additional factors such as mental health, motivation, socio-economic background, and types of learning resources used.

References

Abbott-Chapman, J., Easthope, G., & P O'Connor. (1997). The Influence of Student Gender and Parental Socio-economic Status on Post-school Career Paths. 4, 3–25.

Christiana, O. O. (2014). Institutional Factors Affecting the Academic Performance of Public Administration Students in a Nigerian University. *Public Administration Research*, 3(2). <https://doi.org/10.5539/par.v3n2p171>

Dissanayaka, D. M. N. D., Ediriweera, E. A. I. N., & Rathnayake, R. M. T. N. (2021). Impact of technology readiness, internet self-efficacy, and computing experience on employability among professional accounting students. *Journal of SACFIRE*, 1(1).

Identification of the Factors Affecting the Academic Performance of the Science Undergraduates of the Eastern University, Sri Lanka: A Case Study. (2020). *Journal of Education and Practice*. <https://doi.org/10.7176/jep/11-15-17>

Mandal, A., Ghosh, A., Sengupta, G., Das, N., Mukherjee, S., & Bera, T. (2012). Factors affecting the performance of undergraduate medical students: A perspective. *Indian Journal of Community Medicine*, 37(2), 126. <https://doi.org/10.4103/0970-0218.96104>

Myatt, A., & Waddell, C. (1990). An Approach to Testing the Effectiveness of the Teaching and Learning of Economics in High School. *Journal of Economic Education*, 21(3), 355–363. <https://doi.org/10.1080/00220485.1990.10844681>

Newman-Ford, L., Lloyd, S., & Thomas, S. (2009). An investigation in the effects of gender, prior academic achievement, place of residence, age and attendance on first-year undergraduate attainment. *Journal of Applied Research in Higher Education*, 1(1), 14– 28. <https://doi.org/10.1108/17581184200800002>

Nonis, S. A., & Hudson, G. I. (2006). Academic Performance of College Students: Influence of Time Spent Studying and Working. *Journal of Education for Business*, 81(3), 151–159. <https://doi.org/10.3200/joeb.81.3.151-159>

Nonis, S. A., & Hudson, G. I. (2010). Performance of college students: Impact of study time and study habits. *Journal of Education for Business*, 85(4), 229-238.

Pilcher, J. J., & Walters, A. S. (1997). How Sleep Deprivation Affects Psychological Variables Related to College Students' Cognitive Performance. *Journal of American College Health*, 46(3), 121–126. <https://doi.org/10.1080/07448489709595597>

Rahimpour, M., & Magsoudpour, M. (2011). Teacher-Students' Interactions in Task-Based vs Form-Focused Instruction. *World Journal of Education*, 1(1). <https://doi.org/10.5430/wje.v1n1p171>

Rossi, M. (2017). Factors Affecting Academic Performance of University Evening Students. *Journal of Education and Human Development*, 6(2). <https://doi.org/10.15640/jehd.v6n1a10>

Schmidt, B. (2017). The impact of attendance on college student success: The role of attendance policies and classroom engagement. *Journal of College Student Retention: Research, Theory & Practice*, 19(3), 265-280.

Selwyn, N. (2016). *Education and Technology: Key Issues and Debates*. Bloomsbury Publishing. Win, R., & Miller, P. W. (2005). The Effects of Individual and School Factors on University Students' Academic Performance. *The Australian Economic Review*, 38(1), 1–18. <https://doi.org/10.1111/j.1467-8462.2005.00349.x>

Appendix

Appendix 01

Regression Analysis: GPA versus EngagedClass

Regression Equation

$$\text{GPA} = 1.752 + 0.3347 \text{ engaged_class}$$

Coefficients

Term	Coef	SE Coef	95% CI	T-Value	P-Value	VIF
Constant	1.752	0.179	(1.395, 2.109)	9.77	0.000	
engaged_class	0.3347	0.0497	(0.2357, 0.4336)	6.73	0.000	1.00

Model Summary

S	R-sq	R-sq(adj)	PRESS	R-sq(pred)	AICc	BIC
0.446640	35.86%	35.07%	16.9207	32.84%	106.03	112.98

Appendix 02

Pairwise Pearson Correlations

Sample 1	Sample 2	N	Correlation	95% CI for p	P-Value
Study_Hours	RESI	83	-0.110	(-0.318, 0.108)	0.322
Academic_satisfaction	RESI	83	-0.002	(-0.217, 0.214)	0.989
Academic_satisfaction	Study_Hours	83	0.107	(-0.111, 0.316)	0.334

RESI: difference between the actual observed GPA value and the predicted GPA value calculated by the regression model for each student.